

WEEK – 5 TASK REPORT

DBMS & Business Analytics (BA)

DBMS Task

Payment Transaction Management System

BA Task

Dynamic Portfolio Risk Arbitrage & Value-at-Risk (VaR) Monte Carlo Simulator

This report explains the Week-5 DBMS and Business Analytics tasks in simple terms. The DBMS project is about storing and managing customer payment transactions. The BA project is about studying investment risk using data, covariance, Cholesky decomposition, Geometric Brownian Motion, Monte Carlo simulation, and 99% Value-at-Risk.

Purpose of the Report

The aim is to understand both projects clearly, learn their basic working process, and see how databases and analytics can be used for practical business and financial problems.

PAGE 2 – DBMS: PAYMENT TRANSACTION MANAGEMENT SYSTEM

1. Introduction

A Payment Transaction Management System is a database application used to store and manage payment records. Whenever a customer makes a payment, important information such as customer ID, amount, payment mode, date, and transaction status can be stored in the Payment table.

For example, a customer may pay ₹2,500 using UPI. The database can store the customer ID, payment amount, UPI as the payment mode, the payment date, and a status such as Successful or Failed.

2. Objectives

- Create a Payment table.
- Store payment details in an organized form.
- Record successful, failed, and pending transactions.
- Analyze which payment methods customers use most.
- Generate payment transaction reports.

3. Main Payment Details

Field	Simple Meaning	Example
Payment_ID	Unique payment number	P101
Customer_ID	Customer identification number	C101
Amount	Money paid by customer	2500.00
Payment_Mode	Method used for payment	UPI
Payment_Date	Date of transaction	2026-08-27
Status	Result of payment	Successful

4. Payment Modes

Common payment modes include UPI, Debit Card, Credit Card, Net Banking, and Cash. The database can record the mode selected by each customer.

PAGE 3 – DBMS: TABLE CREATION AND TRANSACTIONS

5. Creating the Payment Table

The Payment table stores one row for each transaction. Payment_ID is the primary key, so every payment has a unique identity.

```
CREATE TABLE Payment (  
Payment_ID INT PRIMARY KEY,  
Customer_ID INT,  
Amount DECIMAL(10,2),  
Payment_Mode VARCHAR(30),  
Payment_Date DATE,  
Status VARCHAR(20)  
);
```

6. Inserting Sample Records

```
INSERT INTO Payment VALUES  
(101, 1, 2500.00, 'UPI', '2026-08-27', 'Successful');  
INSERT INTO Payment VALUES  
(102, 2, 1500.00, 'Card', '2026-08-27', 'Failed');
```

7. Managing Transaction Status

A payment can normally be marked as Successful, Failed, or Pending. Successful means the money was received. Failed means the payment did not complete. Pending means the final result is not yet confirmed.

```
UPDATE Payment  
SET Status = 'Successful'  
WHERE Payment_ID = 102;
```

8. Viewing Transactions

To display all payment records:

```
SELECT * FROM Payment;
```

To display only successful payments, the condition Status = 'Successful' can be used. This makes it easy for staff to check transaction results.

PAGE 4 – DBMS: ANALYSIS AND REPORTS

9. Analyzing Payment Methods

The database can be used to find which payment method is most popular. For example, the following query counts the number of transactions for each payment mode.

```
SELECT Payment_Mode, COUNT(*) AS Total_Transactions
FROM Payment
GROUP BY Payment_Mode;
```

10. Finding Successful and Failed Payments

```
SELECT Status, COUNT(*) AS Total
FROM Payment
GROUP BY Status;
```

11. Total Payment Amount

```
SELECT SUM(Amount) AS Total_Amount
FROM Payment
WHERE Status = 'Successful';
```

12. Payment Transaction Report

A report can show payment ID, customer ID, amount, payment mode, date, and status. Such reports help businesses check daily transactions, identify failed payments, and understand customer payment preferences.

13. Benefits

- Faster payment record management.
- Easy identification of failed transactions.
- Better customer and payment analysis.
- Accurate transaction reports.
- Reduced manual record keeping.

DBMS Conclusion

The Payment Transaction Management System provides a simple and organized way to store, update, analyze, and report payment information using a database.

PAGE 5 – BA: PORTFOLIO RISK SIMULATOR

1. Introduction

The Dynamic Portfolio Risk Arbitrage & Value-at-Risk (VaR) Monte Carlo Simulator is a financial analytics project. It studies the possible risk of a portfolio containing many different assets.

The project works with historical financial data and uses mathematical and statistical techniques to create many possible future scenarios. The final goal is to estimate the portfolio's 99% Value-at-Risk.

2. Main Requirements

- Work with a multi-asset portfolio containing up to 500 assets.
- Calculate the historical covariance matrix.
- Perform Cholesky decomposition.
- Create correlated random market movements.
- Simulate future asset prices using Geometric Brownian Motion.
- Calculate portfolio losses and 99% VaR.

3. What is a Portfolio?

A portfolio is a collection of investments. Instead of investing in only one asset, an investor may hold many assets. The risk of the complete portfolio depends on the individual assets and how they move together.

4. What is Risk?

Risk means the possibility that an investment may lose value. In financial markets, prices can move up or down because of market conditions. A risk simulator helps estimate possible losses before making decisions.

PAGE 6 – BA: COVARIANCE AND CHOLESKY DECOMPOSITION

5. Historical Returns

Historical prices are collected for the assets. From these prices, returns are calculated. Returns show how much an asset's price changed over a period.

6. Covariance Matrix

A covariance matrix shows how asset returns move in relation to each other. If two assets usually move in the same direction, their covariance can be positive. If they often move in opposite directions, covariance can be negative.

For 500 assets, the covariance matrix has 500 rows and 500 columns. It therefore contains information about the relationships between all the assets.

Simple Example

Suppose Asset A and Asset B usually increase together. Their covariance may be positive. If Asset A increases while Asset B usually decreases, their covariance may be negative.

7. Cholesky Decomposition

Cholesky decomposition is a mathematical technique used when a covariance matrix is suitable for decomposition. It breaks the covariance matrix into a special triangular matrix and its transpose.

In simple terms, independent random numbers are first generated. Cholesky decomposition is then used to transform them into random movements that follow the relationships between the assets.

Why is it Needed?

Real assets are not completely independent. Their prices can be affected by common market events. Cholesky decomposition helps the simulation represent these relationships more realistically.

PAGE 7 – BA: GBM AND MONTE CARLO SIMULATION

8. Geometric Brownian Motion (GBM)

Geometric Brownian Motion is a mathematical model commonly used to simulate possible future asset prices. It considers the current price, expected return, volatility, time, and random market movement.

In simple terms, GBM asks: 'If the market behaves in different possible ways, what could the asset price look like in the future?'

9. Monte Carlo Simulation

Monte Carlo simulation generates many possible future scenarios. For every scenario, the model creates future prices for all assets and calculates the resulting portfolio value.

Basic Simulation Process

1. Start with current asset prices.
2. Use historical returns and volatility.
3. Generate random market movements.
4. Correlate the movements using the covariance matrix and Cholesky decomposition.
5. Simulate future prices using GBM.
6. Calculate the portfolio value for every scenario.
7. Compare future values with the current portfolio value.
8. Calculate the possible losses.

10. Independent GBM Paths

The simulator can create separate price paths for each asset. The random movements are generated together so that the final paths reflect the relationships between the assets.

Simple Idea

If 10,000 simulations are generated, the system creates 10,000 possible portfolio outcomes. These outcomes form a distribution of possible future gains and losses.

PAGE 8 – BA: 99% VaR, RESULTS AND CONCLUSION

11. Value-at-Risk (VaR)

Value-at-Risk is a risk measurement that estimates how much a portfolio could lose over a selected time period at a particular confidence level.

12. What does 99% VaR mean?

A 99% VaR means that, according to the model, about 99% of simulated outcomes have losses smaller than the VaR level, while about 1% of outcomes are worse. It is an estimate, not a guarantee.

Example: If the calculated one-day 99% VaR is ■1,00,000, the model estimates that a loss larger than ■1,00,000 would occur in roughly 1% of simulated cases under the model assumptions.

13. Complete BA Workflow

Historical Prices → Returns → Covariance Matrix → Cholesky Decomposition → Correlated Random Noise → GBM Paths → Portfolio Values → Losses → 99% VaR

14. Expected Output

- Historical covariance matrix for the assets.
- Correlated random market movements.
- Many possible future price paths.
- Simulated portfolio values and losses.
- Final 99% VaR estimate.

15. Overall Conclusion

The DBMS project teaches how payment information can be stored and analyzed in a structured database. The BA project teaches how financial data can be used to understand portfolio risk. The BA simulator combines historical data, covariance, Cholesky decomposition, GBM, and Monte Carlo simulation to estimate 99% VaR. Together, these tasks demonstrate how technology and data analysis can support business decision-making.

Key Learning

DBMS helps organize and manage data, while Business Analytics helps convert data into useful information for decisions. Both are important skills for modern business and technology applications.